

CarHub Project Summary Notes

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Frontend:

- Created login.html (index.html) and register.html pages with forms.
- Added form validation and fetch API calls to connect to backend routes (/login, /register).
- Implemented a home page with navigation (home, models, services, about, contact).
- Added a 'Book a Test Drive' button to show a form with file uploads (license, aadhar, photo).
- Used JavaScript to toggle form visibility and handle form submission via fetch with FormData.
- Added basic CSS for styling and animations for better UI/UX.
- Ensured responsiveness and usability improvements (sticky navbar, hover effects).

Backend:

- Used Node.js with Express.js framework for the backend server.
- Connected MySQL database to store users and test drive bookings.
- Created API endpoints:
 - POST /register : to register a new user (fullName, email, password).
 - POST /login : to authenticate a user by email and password.
 - POST /book-test-drive : to save test drive details with file uploads.
- Used multer middleware for handling multipart/form-data and saving uploaded files to 'uploads/' folder.
- Setup MySQL tables:
 - users (id, full_name, email, password)
 - test_drives (id, user_id, car_model, quantity, name, email, phone, license_file, aadhar_file, photo_file, b
- Fixed issues with missing columns and file path saving in database.

- Handled CORS and JSON parsing with appropriate middlewares.
- Error handling added for database and server errors.
- Ensured server listens on port 5000.

Setup & Running:

1. Install Node.js, MySQL, and required npm packages (express, cors, mysql2, multer).
2. Create MySQL database and tables as per schema.
3. Create backend folder, add server.js and db.js files.
4. Run `node server.js` to start the server.
5. Open frontend files in browser (login.html, register.html, home.html).
6. Test user registration, login, and booking test drive with file uploads.
7. Check MySQL data to verify saved records.

Notes:

- Make sure uploads/ folder exists in backend directory and has write permissions.
- Keep frontend and backend URLs consistent (e.g., localhost:5000 for API calls).
- Passwords stored as plain text currently - for production, implement hashing.
- Further UI enhancements and backend validations recommended.

This summary covers all discussed topics for your CarHub project frontend and backend integration.